

19/2/26

Some More Fourier Transform

Convolution Theorem

- Convolution in the ~~time~~ ^{large} domain becomes ~~a~~ Multiplication in the Frequency / Space domain

$$\left(\sum_{n=-\infty}^{\infty} \left[\sum_{\tau=-\infty}^{\infty} x(\tau) y(\tau-n) \right] e^{-j\omega \tau} \right) = X(\omega) \cdot Y(\omega)$$
$$X(n) \Rightarrow \boxed{F(x * y) = F(x) \cdot F(y)}$$

Note that we are writing this in the discrete form as in most cases in data analysis, we only have access to a specific set of values.

Similarly,

$$\boxed{F(x, y) = F(x) * F(y)}$$

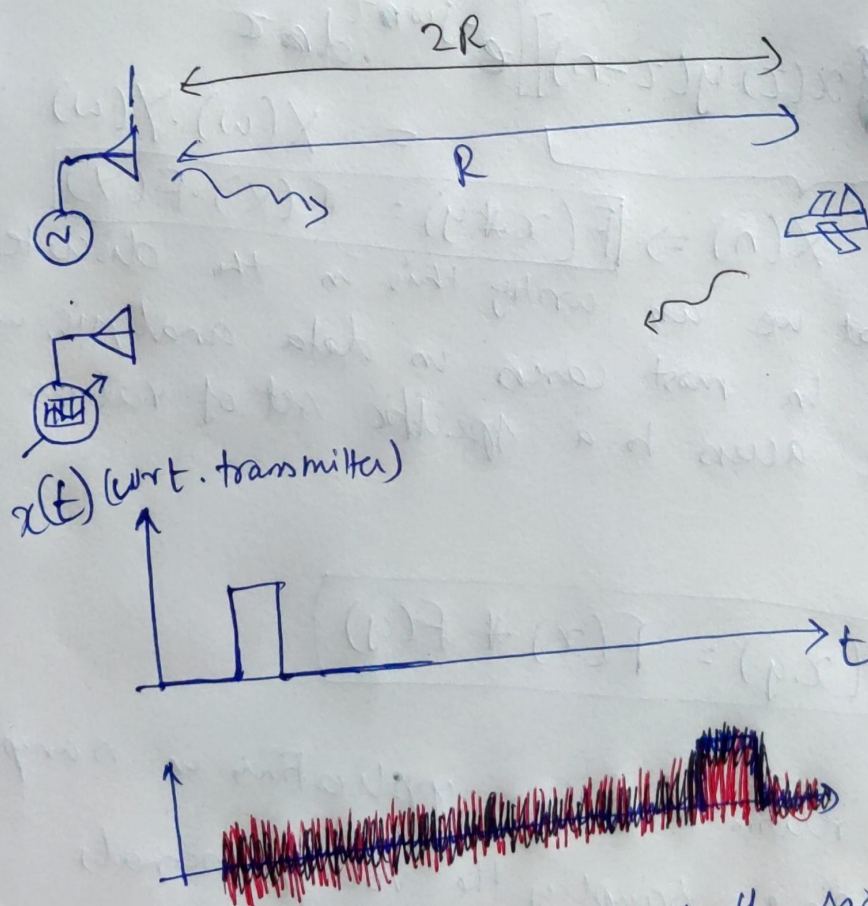
Important relation that has applications in sampling the u.v. plane, sampling the voltage signals appropriately etc.

Now, we will try and understand the "Course" of Convolution, the "Correlation".

In signal processing correlation is used to get the similarity of a "signal" in another.

Ex: radar, a pulse is correlated with a delayed version.

We will use this to motivate the operation of correlation.



We want to try and see if ~~the~~ the signal that we sent $x(t)$ is contained in what we received, which is $y(t)$.

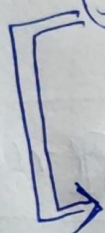
The signal that we sent is very well known to us.
Can we use this to detect the reflected signal?

Steps to do it:

① Start a timer when the first signal leaves

② Keep the receiver open for some time (t)

③ Keep comparing the received signal with the template of the transmitted ~~reflected~~ signal.



↓
But ~~what's coming~~
We turn to our "old friend" convolution for this.

But the transmitted signal, as we represent will be flipped, as the first time is what would show up at the receiver [if at all the signal is present]

if $s(t)$ was the Tx pulse, $r(t+\tau)$ was the reflected pulse, that comes after a delay τ .

Then,

$$\text{Sim}(\tau) \sim \int s(t) \cdot r(t+\tau) dt$$

Note that that this axis is the delay axis; and not time, like in convolution

→ If I want to compare the template, then

$s(t) \sim h(-t)$ a filter kernel defined using the pulse

then,

$$\boxed{\text{Sim}(z) = [r * s(-t)](z)}$$



Matched Filter.

This is called correlation. ~~which is for correlation~~

We can also define a convolution theorem; as an extension of the convolution.

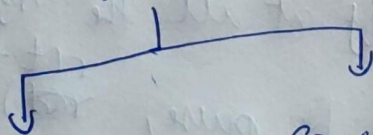
~~$F\{s(t) \cdot r(t)\}$~~

$$\boxed{F\{s(t) \cdot r(t+z)\} = F^*[s(t)] \cdot F[r(t)]}$$



Correlation theorem

Correlation



Auto



Correlation of the signal with itself.

Cross



Correlation of signal with another

like in radar ex.

In radio astronomy we use both flavors of correlation

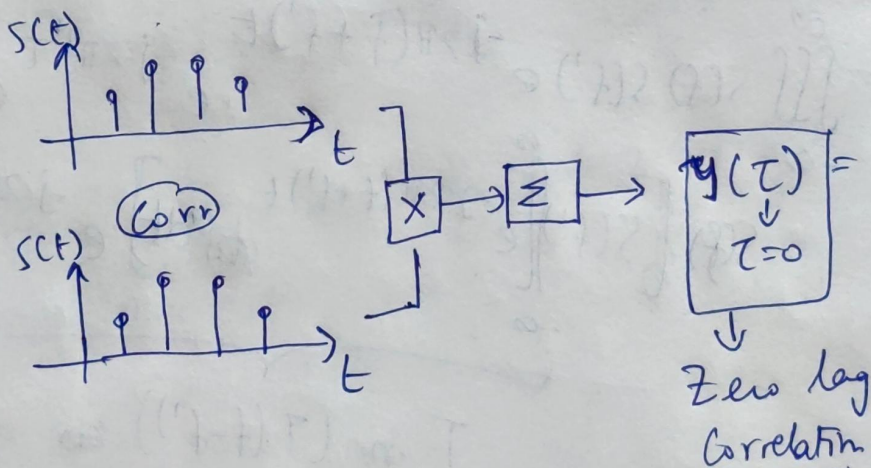
- Auto correlation is usually used to determine the

"Power spectrum" of a ~~obj~~ source/signal.

↓
Distribution of power across freq

From the correlation equation, Auto correlation can be written as:

$$S(t) \cdot S(t+\tau)$$



we will change values of τ as we find ~~not~~ fit.

If we take an FFT of the correlation, then,

$$\int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} S(t) \cdot S(t+\tau) dt \right] e^{-j\omega\tau} d\tau$$

$$\Rightarrow \frac{dS}{d\tau} = \dots$$

$$S(t) \rightarrow \int_{-\infty}^{\infty} S(f) e^{-j2\pi f t} df$$

$$S(t+\tau) \rightarrow \int_{-\infty}^{\infty} S(f') e^{-j2\pi f' (t+\tau)} df'$$

Sub this in the Auto corr. integral, then,

$$r_{xx}(\tau) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} S(f) e^{-j2\pi f t} df \int_{-\infty}^{\infty} S(f') e^{-j2\pi f' (t+\tau)} df'$$

Group exponentials

$$r_{xx}(\tau) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} S(f) e^{-j2\pi f t} S(f') e^{-j2\pi f' (t+\tau)} df df' dt$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} S(f) S(f') e^{-j2\pi (f+f') t} e^{-j2\pi f' \tau} df df' dt$$

$$= \int_{-\infty}^{\infty} S(f) \left[\int_{-\infty}^{\infty} S(f') e^{-j2\pi (f+f') t} dt \right] e^{-j2\pi f' \tau} df'$$

$$T \sin(T(f-f'))$$

$$\text{as } T \rightarrow \infty, T\delta(f-f')$$

$$\Rightarrow r_{xx}(\tau) = \int_{-\infty}^{\infty} S(f) S(f') T\delta(f-f') df' df$$

$$\text{But } \int_{-\infty}^{\infty} S(f') \delta(f-f') df' = S(f)$$

Shifting property.

$$\therefore r_{xx}(\tau) = \int S(f) \cdot S(f) e^{-j2\pi f\tau} df$$

$$r_{xx}(\tau) = \int_{-\infty}^{\infty} \underbrace{|S(f)|^2}_{\text{PS}} e^{-j2\pi f\tau} df$$

$$r_{xx}(\tau) = \int_{-\infty}^{\infty} \text{PS} e^{-j2\pi f\tau} df$$

Take ifft, then,

$$\left[r_{xx}(\tau) e^{j2\pi f\tau} d\tau \right] = \text{PS}$$

$\therefore$ Power spectrum is the auto correlation's fourier transform.

$$\text{PS} \rightarrow \frac{\text{W}}{\text{Hz}}$$

also called power spectral density.

for complex sequences, PS will be

$$\text{PS} = \frac{|S(\omega) S^*(\omega)|}{(\text{Magnitude})^2}$$